



LA GRANDEE INTERNATIONAL COLLEGE

Simalchaur, Pokhara, Nepal

A Project Proposal

on

“Hammro School”

Submitted To

LA GRANDEE International College

Bachelor of Computer Application (BCA) Department

In partial fulfilment of the requirements for the degree of

Bachelor of Computer Application under Pokhara University

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1. Introduction

Education is one of the most essential pillars of society, and managing school operations effectively is critical for ensuring quality learning outcomes and institutional growth. Traditional methods of school administration, which rely heavily on manual record-keeping, are prone to human errors, data redundancy, and operational inefficiency.

The Hammro School is a software-based solution designed to automate and digitize the management of school-related operations including student registration, attendance tracking, marks management, class scheduling, and report generation. The system aims to provide an efficient, reliable, and user-friendly platform for schools to manage their academic and administrative operations seamlessly.

This project is developed as part of the Bachelor of Computer Application (BCA) program under Pokhara University. The system will be a desktop application providing a centralized database to manage all aspects of school management from student enrollment to final result generation.

2. Problem Statement

The primary challenge in traditional school administration is the dependence on manual and fragmented management processes that reduce efficiency, accuracy, and security of academic records. Schools handle large amounts of student, teacher, and examination data daily, but the lack of a centralized digital system creates delays, inconsistencies, and security risks. The key problems addressed by this project are as follows:

- Manual entry of student data leads to frequent calculation errors and data inaccuracies in marks, grades, and examination results.
- No real-time visibility of student attendance causes delays in notifying parents and makes it difficult to track chronic absenteeism.
- Lack of centralized student and teacher management makes tracking academic records, staff details, and student performance difficult.
- Generating report cards and performance summaries manually is time-consuming, repetitive, and unreliable.
- Absence of proper access control allows unauthorized personnel to alter, delete, or misuse important academic records.
- Data is scattered across multiple files and registers, making it difficult to audit, review, or verify examination results efficiently.
- There is a high risk of data loss due to damage, fire, theft, or misplacement of physical school records.
- This project aims to address these issues by developing a Hammro School using modern digital technologies to improve efficiency, accuracy, security, and centralized data management.

3. Objectives

The following are the key objectives of the Hammro School:

- To develop a centralized digital platform for recording and managing student academic data and school operations.
- To automate attendance tracking and generate real-time academic performance reports.
- To maintain records of students, teachers, classes, and subjects efficiently.
- To reduce manual errors and improve data accuracy in school administration.
- To provide a role-based login system ensuring data security.
- To generate mark sheets, report cards, and performance summaries for academic decision-making.

4. Methodology

The project will be designed according to the format prescribed by the college. It will be developed using Java programming language and JavaFX GUI toolkit. The Agile software development methodology will be used as the program development process.

Agile is an iterative software development approach where the project is developed in small phases called sprints. Each sprint focuses on developing and testing specific modules of the Hammro School. This methodology is chosen because it supports continuous improvement, early error detection, flexibility, and timely project completion.

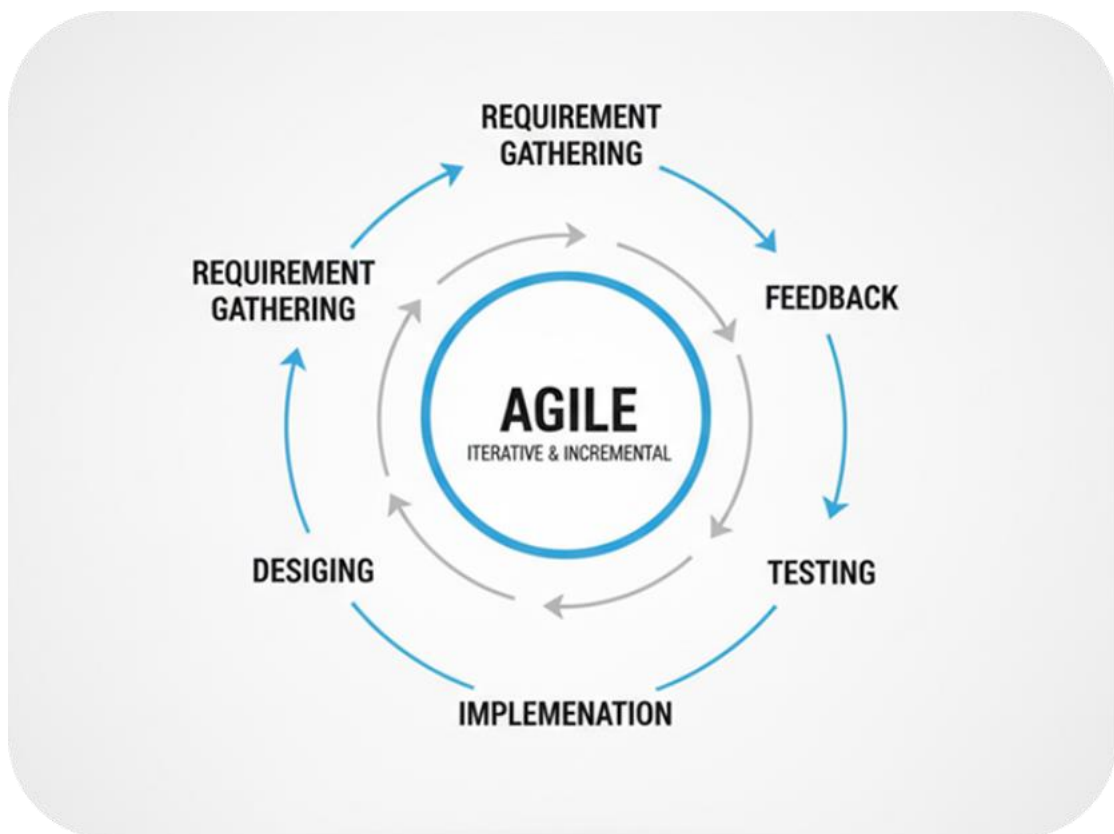


Figure 4.1: Agile Software Development Model

5. Gantt Chart

The Hammro School uses an Agile model, developing the project through iterative weekly sprints over 45 days. This approach allows for continuous testing and flexible adaptation from May to June. Three sprint cycles cover core setup, academic modules, and final integration. This timeline ensures that design diagrams and database schemas are finalized before the final release. Testing and documentation run concurrently to ensure a reliable, secure system for school administration.

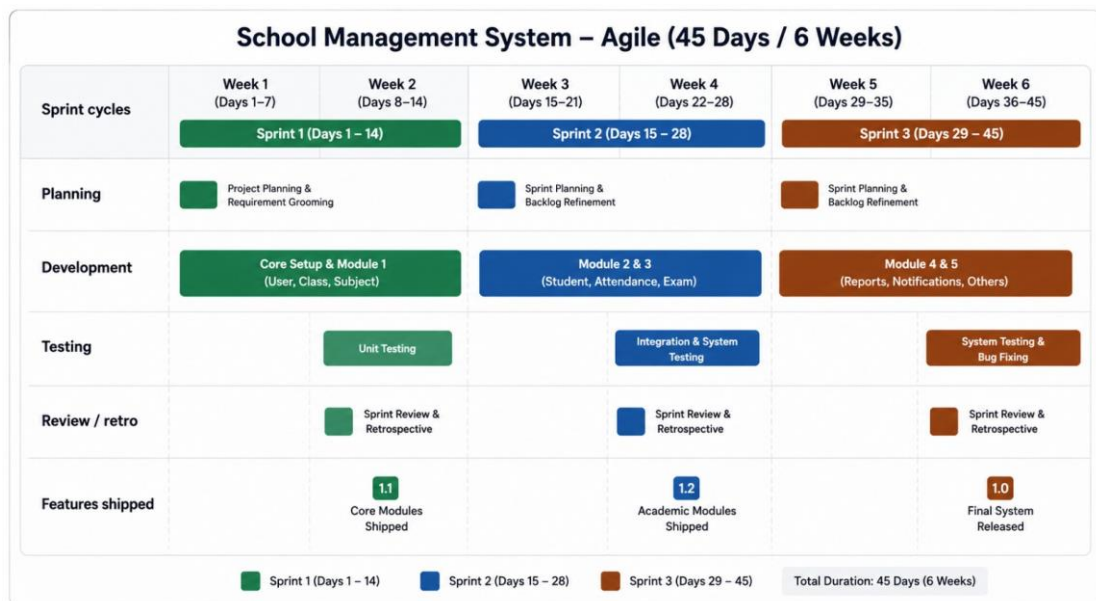


Figure 5.1: Project Gantt Chart (May – July 2026)

6. Deliverables

Deliverables refers to the actual files that we submit with our project, described in the following list:

- A single compressed archive (.rar) containing all project files, including the complete Java source code, the Bouncy Castle dependency, build scripts, and any required resource files.
- A user manual (README.md) that details command line usage and an executable file (.exe) for execution of the application.

7. References

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- Schwaber, K., & Sutherland, J. (2023). *The Scrum Guide: The Definitive Guide to Scrum: The Rules of the Game*. Scrum.org.
- **Scene Builder:** Used for designing FXML-based user interface layouts in a visual and structured way.
- **JDBC (Java Database Connectivity):** Acts as the communication bridge between the JavaFX application and the SQL Server database, enabling data retrieval, insertion, updating, and deletion.
- **T-SQL (Transact-SQL):** Used for writing SQL Server-specific queries, stored procedures, and database-side logic to improve performance and maintain data integrity.